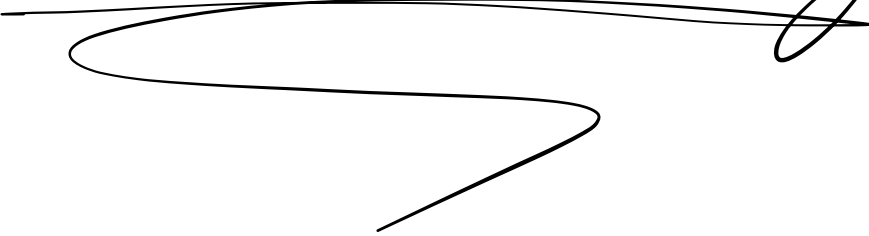


Hierarchical
Modeling



Comparing Multiple Related Groups

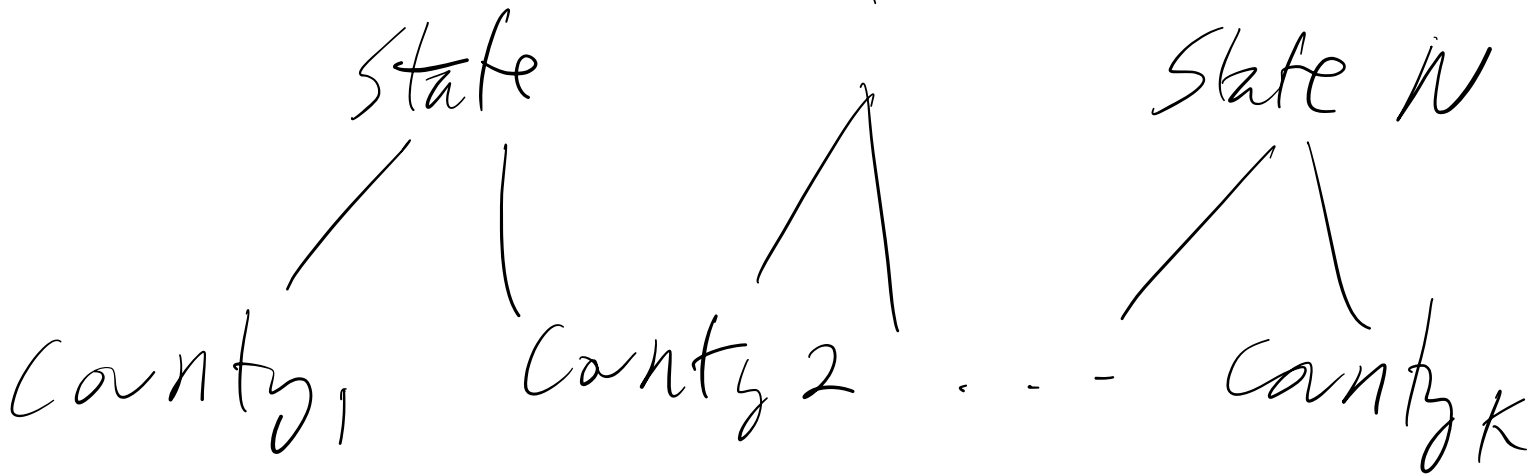
- Hierarchy of nested populations
- Models which account for this are called *hiearchical* or *multi-level* models

Some examples:

- Patient outcomes within several different hospitals
- People within counties in the United States (e.g. Asthma mortality example)
- Athlete performance in sports
- Genes within a group of animals

Example

U.S.



Eight schools example

- A study was performed for the Educational Testing Service (ETS) to evaluate the effects of coaching programs on SAT preparation
- Each of eight different schools used a short-term SAT prep coaching program
- Compute the average SAT score in those who did take the program minus those that did not participate in the program
- We observe the average difference varies by school. What accounts for these differences?

- Economic disparities
- Student access to other programs.
- Student- to - teacher ratio
- Students Baseline.

Eight schools example

- Interested in "real" differences due to training
- Want to reduce effect of chance variability
- How do we estimate the effect of the program in each of the schools?
- Two extremes:

- Estimate the effect of the program in every school independently

- A separate prior distribution for each school effect

- Or assume the effect is the same in every school

- Combine all the data

- A compromise between the above 2 options?

Students are
randomized
into program.

"No Pooling"

"Complete
pooling"

More
Data
"pooling"

"partial Pooling"

Eight Schools Example

observed difference:
training minus baseline.

$y_j \sim N(\theta_j, \sigma_j^2)$

sampling variance (known)
True Difference

- y_j is the observed effects of the program in school j
 - Based on a sample of test scores from those in the program and those not in the program
- θ_j are the true *unknown* effects of the program in school j
- Variances, σ_j^2 , are *known*
 - Determined by the number of students in the sample

$$\sigma_j^2 = \frac{\sigma^2}{n_j}$$

Eight Schools Example

```
J <- 8
y = c(28, 8, -3, 7, -1, 1, 18, 12)
sigma <- c(15, 10, 16, 11, 9, 11, 10, 18)
```

Actual y_j 's.

- Assuming the effect of the program on each school is identical. $\theta_j = \theta_k$ $\forall j, k$
- What are the chances of seeing a value as large as 28?
- As small as -3?

$$y_j \sim N(\theta, \sigma_j^2)$$

$$\theta_j = \theta$$

(same for all schools)

Comp $\hat{\theta}_{MLE}$
pooling

$\hat{\theta}_{MLE}$:

$$\frac{\sum_{j=1}^8 \frac{1}{\sigma_j^2} y_j}{\sum_{j=1}^8 \frac{1}{\sigma_j^2}}$$

$$\approx 7.7$$

(Precision-weighted
Average)

Bias-Tradeoff.

Eight Schools Example

If effects are actual equal, what is it?

```
## Compute the precision from each school  
prec <- 1/sigma^2
```

```
## global estimate is a weighted average  
mu_global <- sum(prec * y / sum(prec))  
mu_global
```

Precision weighted.

```
## [1] 7.685617
```

*⊖ assuming effectiveness
is identical across all schools.*

Eight Schools Example

- Assume the effect of the program on each school is identical, i.e.

$$\theta_j = \mu \approx 7.7$$

- What are the chances of seeing a value as large as 28?
- As small as -3?

```
# 1000 datasets with mean mu_global but different sigmas
```

```
## Chance of seeing a value greater than 28
```

```
mean(sapply(1:1000, function(x)  
  max(rnorm(J, mean=mu_global, sd=sigma))) >= 28)
```

```
## [1] 0.348
```

$$\hat{=} \Pr(\max(Y_1, \dots, Y_8) \geq 28 \mid \theta = 7.7)$$

```
## Chance of seeing a value less than -3
```

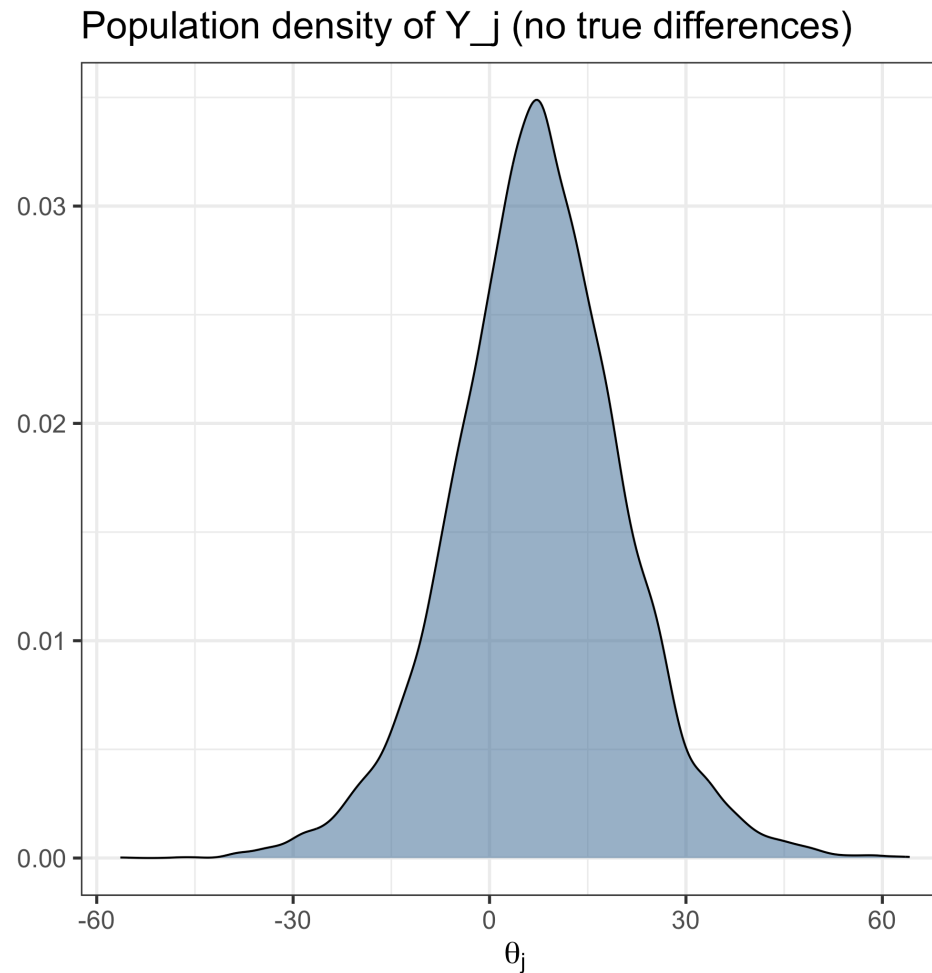
```
mean(sapply(1:1000, function(x)  
  min(rnorm(J, mean=mu_global, sd=sigma))) <= -3)
```

```
## [1] 0.816
```

$$\Pr(\min(Y_1, \dots, Y_8) \leq -3 \mid \theta = 7.7)$$

Eight Schools Example

Density of $Y_j \sim N(\theta_{\text{pooled}}, \sigma_j^2)$



Eight Schools Example

$$y_j \sim N(\theta_j, \sigma_j^2)$$

- θ_j are the true unknown effects of the program in school j
- y_j is the observed effects of the program in school j
 - Based on a sample of test scores from those in the program and those not in the program
 - Number of people in the sample determine the magnitude of σ_j^2

Eight Schools Example

- How do we estimate θ_j ?

(8 separate problems)

- Independent: $\hat{\theta}_j^{(MLE)} = y_j$ is the MLE

Indep. $\hat{\theta}_{j,MLE} = y_j$

Fully Pooled: $\hat{\theta}_{j,MLE} = \hat{\theta} =$

- Identical effects: $\hat{\theta}_j^{(pool)} = \frac{\sum_i \frac{1}{\sigma_i^2} y_i}{\sum_i \frac{1}{\sigma_i^2}} = \theta^{(pool)}$ for all j

(no pooling)

(complete pooling)

- Same effect for all schools: estimate using a weighted average of the observed effects

partial pooling.

$$\hat{\theta}_j^{(shrink)} = w \hat{\theta}_j^{MLE} + (1-w) \hat{\theta}_j^{pooled.}$$

Think of as data informed prior mean

Reminder: $y_j \sim N(\theta_j, \sigma_j^2)$

$\theta_j \sim N(\mu, \tau^2)$

$$E(\theta_j | y_j) = w y_j + (1-w) \mu$$

prior
Guess

$$w = \frac{1/\sigma_j^2}{1/\sigma_j^2 + 1/\tau^2}$$

Full Hierarchical

Also assumes

μ is unknown
and learn it.

$\mu \sim$ Prior on Global Mean (Hyperprior)

$\theta_j \sim N(\mu, \tau^2)$ (prior for each group mem)

$y_j \sim N(\theta_j, \sigma_j^2)$

(Sampling)

$P(\mu, \theta_1, \dots, \theta_g | y_1, \dots, y_g)$

Eight Schools

```
theta_j_mle <- y  
theta_j_mle
```

No Pooling

```
## [1] 28 8 -3 7 -1 1 18 12
```

θ_j

```
theta_j_pooled <- rep(sum(1/sigma^2 * y) / sum(1/sigma^2), J)  
theta_j_pooled
```

```
## [1] 7.685617 7.685617 7.685617 7.685617 7.685617 7.685617 7.685617 7.685
```

Complete Pooling.

Is there a middle ground between two extremes?

$$y_j \sim N(\theta_j, \sigma_j^2)$$

$$\theta_j \sim N(\mu, \tau^2)$$

shared Prior
For all
schools.

Also be unknown.

$$\mu \sim N(\mu_0, \tau_0^2)$$

$$\underline{P(\mu, \theta_1, \theta_2, \dots, \theta_8 / y_1, \dots, y_8, \sigma_1, \dots, \sigma_8)}$$

Why Shrink?

$$y_j = \theta_j + \epsilon_j$$

↑
signal

↑
noise.

$$\theta_j \sim N(\mu, \tau^2)$$

$$\epsilon_j \sim N(0, \sigma_j^2)$$

$$\text{Var}(Y) = \text{Var}(\theta + \epsilon) = \text{Var}(\theta) + \text{Var}(\epsilon)$$

$$= \tau^2 + \sigma^2$$

Variance of y 's is too big.

Signal variance is τ^2 .

Eight schools example

$$E[\theta_j | y_j, \mu] =$$

A hierarchical model:

$$\theta_j \sim N(\mu, \tau^2)$$

$$y_j \sim N(\theta_j, \sigma_j^2)$$

$$w y_j + (1-w) \mu$$

$$w = \frac{1/\sigma_j^2}{\frac{1}{\sigma_j^2} + \frac{1}{\tau^2}}$$

- Add a *shared* normal prior distribution to θ_j
- Assume the global mean, μ is also unknown
- How do we choose prior for μ ?

$$\mu \sim N(\mu_0, \tau_0^2)? \text{ or } p(\mu) \propto 1?$$

Improper prior.

- Need to estimate all of $(\mu, \theta_1, \dots, \theta_8)$ with MCMC
- τ^2 determines how much weight we put on the independent estimate vs the pooled estimate

proper

- Compromise: $\hat{\theta}_j^{(shrink)} = w\theta_j^{(MLE)} + (1 - w)\theta_j^{(pool)}$
- How is this different from the standard normal-normal model we saw before?

Intuition behind shrinkage

- $Y_j = \theta_j + \epsilon_j$ and for simplicity assume that the variance of ϵ , σ^2 for all j

- θ_j represents true differences between schools (signal)
- ϵ_j is sampling variability (noise, chance variation)

- $Var(Y_j) = Var(\theta_j) + Var(\epsilon_j) = \tau^2 + \sigma_j^2$

Signal variance + Noise

- The variance of the observed outcomes is the sum of signal variance, τ^2 , and the sampling variance σ_j^2

- Consequence: the observed outcomes always have higher variance across groups than the signal

- $Var(Y_j) > Var(\theta_j)$ $= \tau^2$ *(true variation in effectiveness across schools)*

- Intuition: reduce the variance by shrinking Y_j 's closer together!

- Want the variance of the shrunken estimates to be close to τ^2

Eight Schools examples

Comments:

- The global average, μ , is a parameter so also has uncertainty

- How do we determine how much to shrink, e.g. how do we determine τ^2 ?

- Is the training program effective in school j ?

- What is $P(\theta_j > 0 \mid y)$? $\approx$ frac of θ_j samples > 0 .

- On average (over all schools) is the training program effective?

- What is $P(\mu > 0 \mid y)$? $\approx$ frac of μ samples > 0 ,

Is t.p. more effective in school 1 or 2?

$$P(\theta_1 > \theta_2 \mid y_1, \dots, y_8) \approx \dots$$

Eight schools example

- If τ^2 is large, the prior for θ_j is not very strong
 - If $\tau^2 \rightarrow \infty$ equivalent to the no pooling model
- If τ^2 is small, we assume a priori that θ_j are very close
 - if $\tau^2 \rightarrow 0$ equivalent to the complete pooling model, $\theta_j = \mu$

Estimating parameters

Metropolis-Hastings

- MH: Need to generate a proposal from a 9-dimensional posterior distribution

- Eight parameters for θ_j and one for μ

$$(\theta_1^* \dots \theta_8^*, \mu^*) \sim N(\theta_1^t, \dots, \theta_8^t, \mu^t)$$

- Gibbs: sample each of the 9 parameters from the complete conditionals

~~Iterations~~

- Sample $p(\theta_j \mid \theta_{-j}, \mu)$
- Sample $p(\mu \mid \theta_1, \dots, \theta_8, \mu)$

- Stan

(HMC)

Eight Schools Estimation

```
J <- 8  
y = c(28, 8, -3, 7, -1, 1, 18, 12)  
sigma <- c(15, 10, 16, 11, 9, 11, 10, 18)  
  
## fix the variance of the prior to a number  
tau <- 20
```

Eight Schools in Stan

// saved as 8schools.stan

data {

int<lower=0> J;

// number of schools

array[J] real y;

// estimated treatment effects

array[J] real<lower=0> sigma; // standard error of effect estim

real<lower=0> tau; // standard deviation in treatment effect

}

parameters {

real mu; μ

// population treatment effect

vector[J] eta;

// unscaled deviation from mu by school

}

transformed parameters {

vector[J] theta = mu + tau * eta;

// school treatment effect

}

model {

target += normal_lpdf(eta | 0, 1);

// prior log-density

y ~ normal(theta, sigma); // log-likelihood

}

$\eta \sim N(0, 1)$

$$\theta_j = \mu + \tau \eta_j$$

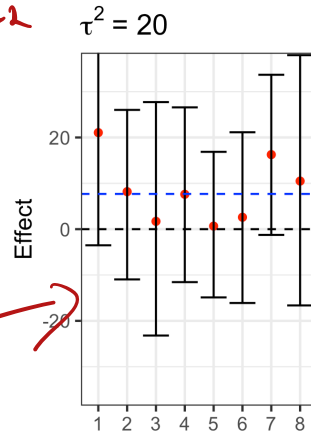
$$\eta_j \sim N(0, 1)$$

$$\Rightarrow \theta_j \sim N(\mu, \tau^2)$$

Eight Schools example

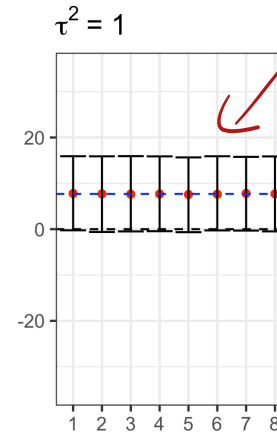
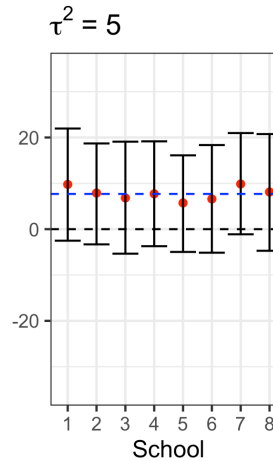
$$w = \frac{1/\sigma_j^2}{1/\sigma_j^2 + 1/\tau^2}$$

Less variation in θ_j 's



$$w \approx 1$$

(Close to no pooling)

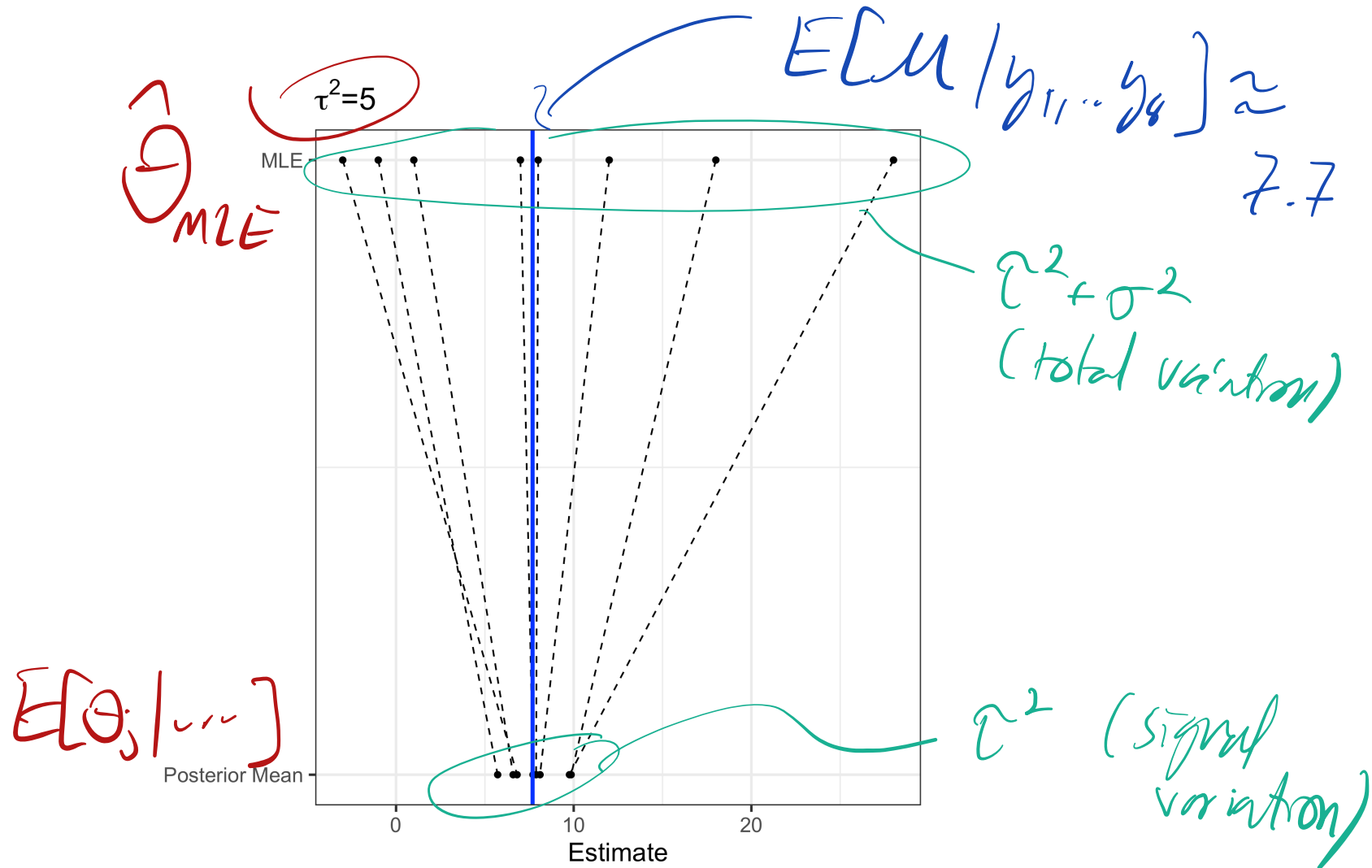


$$w \approx 0$$

(Close to complete pooling)

smaller intervals

MLE vs Posterior Mean



$$(\mu, \tau^2) \sim \text{Hyperprior}$$

$$\theta_j \sim \mathcal{N}(\mu, \tau^2)$$

$$y_j \sim \mathcal{N}(\theta_j, \sigma_j^2)$$

$$P(\tau^2, \mu, \theta_1, \dots, \theta_8 / y_1, \dots, y_8)$$



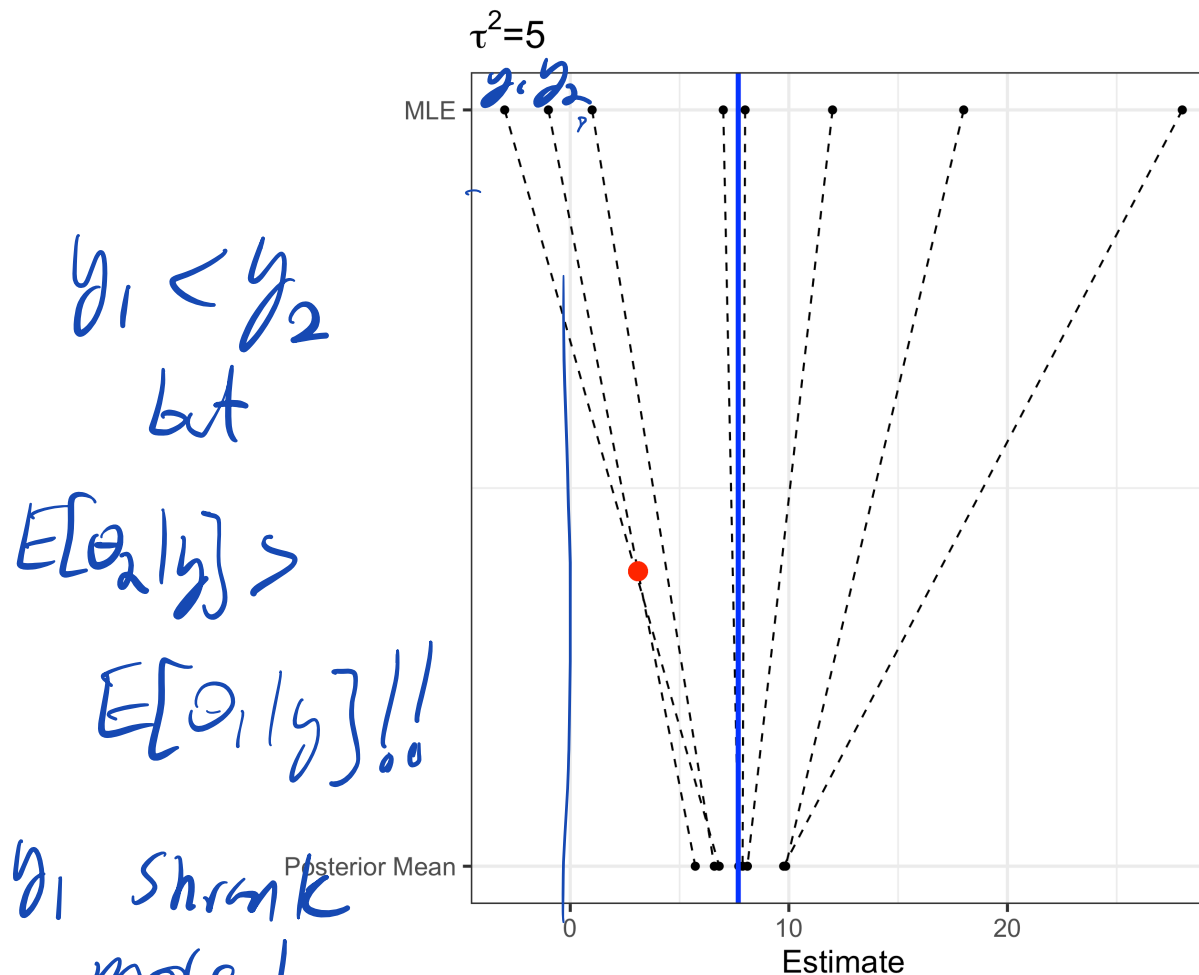
$$y_i = \theta_i + \epsilon_i$$

"sig." "noise."

$$\text{sig var} = \tau^2$$

$$\text{noise var} = \sigma_j^2$$

MLE vs Posterior Mean



$$y_i \sim \mathcal{N}(\theta_i, \sigma_i^2)$$

Why: $\theta_1^2 > \sigma_2^2$

$$E[\theta_i|y] = w_i y_i + (1-w_i)\mu$$

$$w_i = \frac{1/\sigma_i^2}{1/\sigma_i^2 + 1/\tau^2}$$

$$\Rightarrow w_1 < w_2$$

$$1/\sigma_j^2 + 1/\tau^2$$

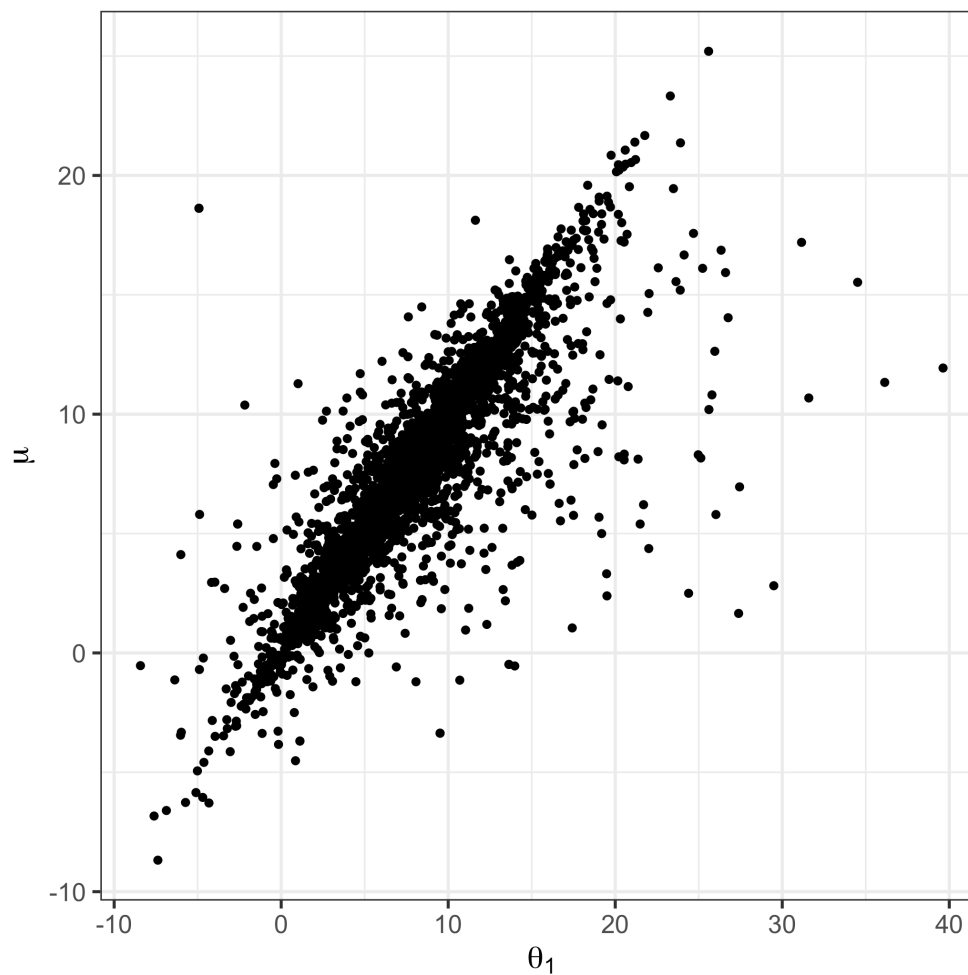
Eight Schools in Stan

```
// saved as eight_schools2.stan
data {
  int<lower=0> J;           // number of schools
  array[J] real y;         // estimated treatment effects
  array[J] real<lower=0> sigma; // standard error of effect estimates
}
parameters {
  real mu;                 // population treatment effect
  vector[J] eta;           // unscaled deviation from mu by school
  real<lower=0> tau;        // standard deviation in treatment effects
}
transformed parameters {
  vector[J] theta = mu + tau * eta; // school treatment effects
}
model {
  tau ~ cauchy(0, 1);
  target += normal_lpdf(eta | 0, 1); // prior log-density
  target += normal_lpdf(y | theta, sigma); // log-likelihood
}
```

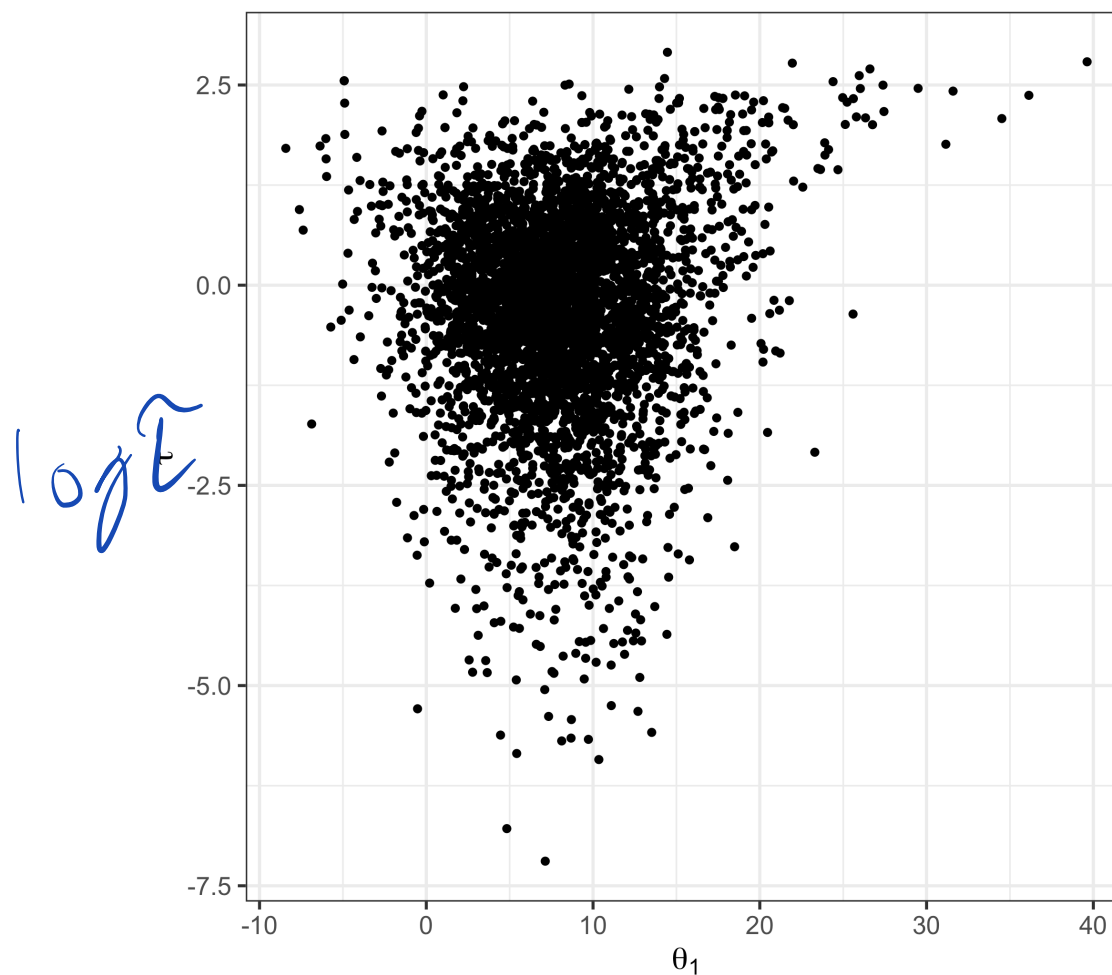
MLE vs Posterior Mean

Posterior mean of $\tau^2 = 1.4287278$

Eight Schools Scatter Plot: θ_1 vs μ



Eight Schools Scatter Plot: θ_1 vs τ

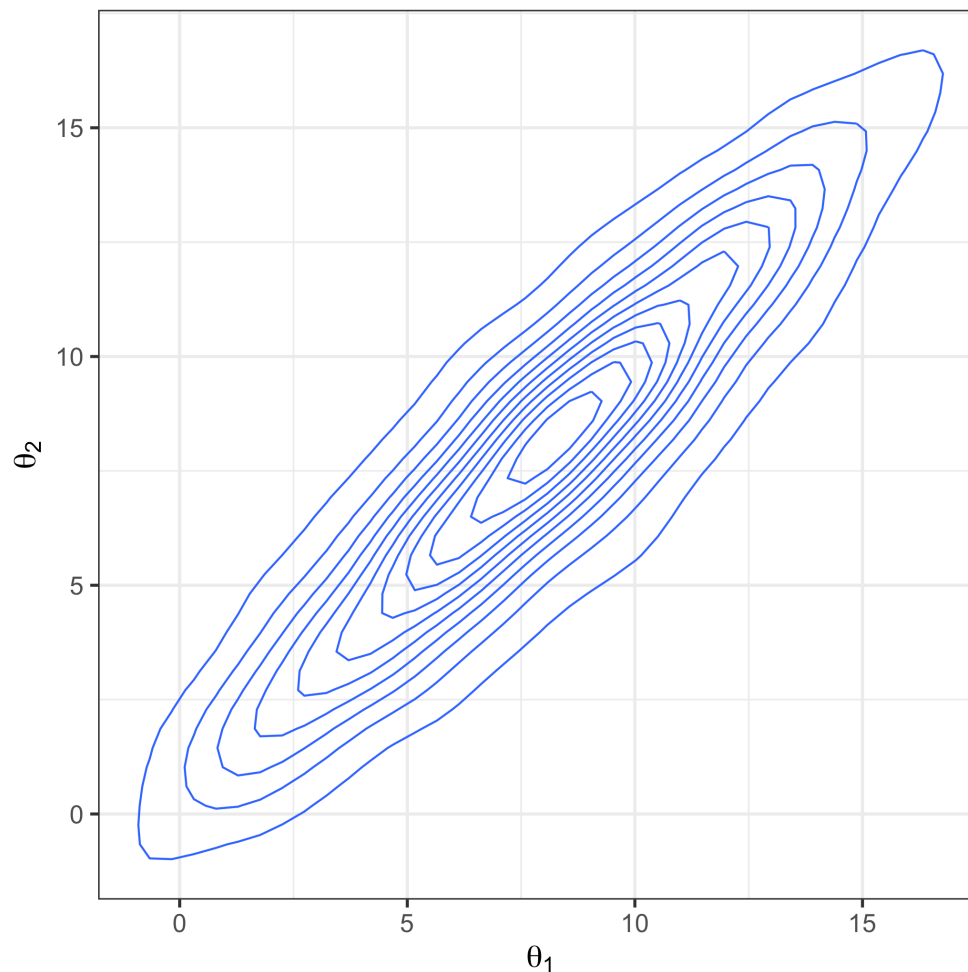


Eight Schools and Independence

- The effects θ_j in each school are *not* independent
- E.g. if we know θ_1 is really large, it means θ_j is also more likely to be large
- However, before we see data the distributions of θ_j 's, are indistinguishable
 - Don't know which θ 's will be large and which small

Eight Schools and Independence

```
## Don't know how to automatically pick scale for object of type  
## <draws_matrix/draws/matrix>. Defaulting to continuous.  
## Don't know how to automatically pick scale for object of type  
## <draws_matrix/draws/matrix>. Defaulting to continuous.
```

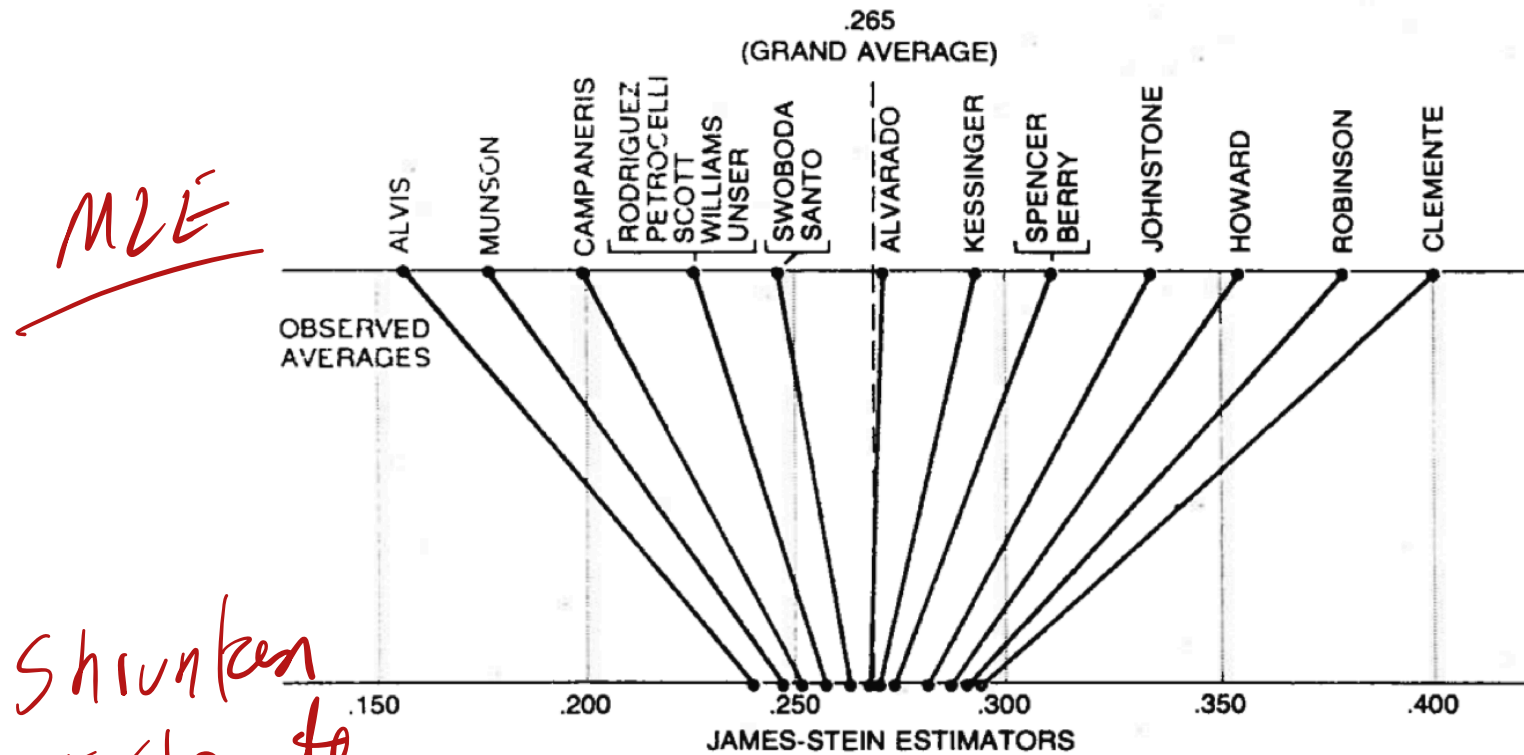


Hierarchical modeling in sports

1. 1970 Batting Averages for 18 Major League Players and Transformed Values X_i , θ_i

i	Player	$Y_i = \text{batting average for first 45 at bats}$	$p_i = \text{batting average for remainder of season}$	At bats for remainder of season	X_i	θ_i
		(1)	(2)	(3)	(4)	(5)
1	Clemente (Pitts, NL)	.400	.346	367	-1.35	-2.10
2	F. Robinson (Balt, AL)	.378	.298	426	-1.66	-2.79
3	F. Howard (Wash, AL)	.356	.276	521	-1.97	-3.11
4	Johnstone (Cal, AL)	.333	.222	275	-2.28	-3.96
5	Berry (Chi, AL)	.311	.273	418	-2.60	-3.17
6	Spencer (Cal, AL)	.311	.270	466	-2.60	-3.20
7	Kessinger (Chi, NL)	.289	.263	586	-2.92	-3.32
8	L. Alvarado (Bos, AL)	.267	.210	138	-3.26	-4.15
9	Santo (Chi, NL)	.244	.269	510	-3.60	-3.23
10	Swoboda (NY, NL)	.244	.230	200	-3.60	-3.83
11	Unser (Wash, AL)	.222	.264	277	-3.95	-3.30
12	Williams (Chi, AL)	.222	.256	270	-3.95	-3.43
13	Scott (Bos, AL)	.222	.303	435	-3.95	-2.71
14	Petrocelli (Bos, AL)	.222	.264	538	-3.95	-3.30
15	E. Rodriguez (KC, AL)	.222	.226	186	-3.95	-3.89
16	Campaneris (Oak, AL)	.200	.285	558	-4.32	-2.98
17	Munson (NY, AL)	.178	.316	408	-4.70	-2.53
18	Alvis (Mil, NL)	.156	.200	70	-5.10	-4.32

Hierarchical modeling in sports

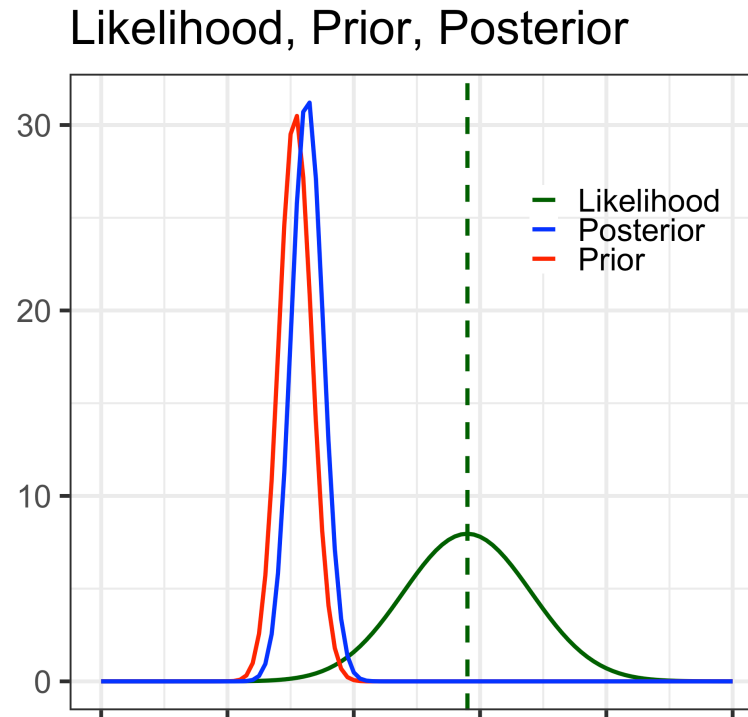


JAMES-STEIN ESTIMATORS for the 18 baseball players were calculated by “shrinking” the individual batting averages toward the overall “average of the averages.” In this case the grand average is .265 and each of the averages is shrunk about 80 percent of the distance to this value. Thus the theorem on which Stein’s method is based asserts that the true batting abilities are more tightly clustered than the preliminary batting averages would seem to suggest they are.

Basketball Example

After 100 shots Robert Covington's 3PT% was 0.49

```
## Warning: A numeric `legend.position` argument in `theme()` was deprecated  
## 3.5.0.  
## i Please use the `legend.position.inside` argument of `theme()` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



Basketball Example

